



# The evolution of election forecasting models in the UK

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## ABSTRACT

This article reviews and analyzes the historical development, innovations, and challenges of election forecasting models in Britain. To anticipate challenges for forecasters, we proceed chronologically from Whiteley's (1979) popularity model and Mughan's (1987) political-economy model to the most recent forecasts in 2019. We then move on to discuss the desired characteristics of forecasts and assess the accuracy of ex-ante UK forecasts. This analysis reveals that while forecasters have had successes, certain elections have posed challenges. In 2015, the models failed to predict a Tory victory, and the snap elections in 2017 and 2019 left scholars with little time to generate and publish forecasts. We discuss the lessons learned and next steps for UK forecasters. And, throughout our article, we highlight the contributions of Harold Clarke who is remembered in this *Electoral Studies* symposium.

## 1. Introduction

Election prediction has a long history in democratic societies. In the early days, prognosticators used bellwethers, straw polls, and other non-causal coincidences to predict the winner (Lewis-Beck and Rice 1992). The emergence of modern public opinion polling and survey methods in the mid-20th century opened up new opportunities for scholars to understand voters and the forces behind electoral victories. Eventually, research on government popularity, the reward-punishment theory, and economic voting allowed election scholars to turn their explanatory theories and empirical findings into predictive models [For introductory discussion and review, see Clarke et al., 2004; Lewis-Beck and Stegmaier, 2000; Lewis-Beck and Stegmaier 2014; Sanders 2000, 2003a; Stegmaier and Norpoth 2017; Whiteley, 1986.].

Model-based election forecasts developed in tandem in the US and UK starting in the late 1970s and 1980s, thanks to the growing body of electoral research there (Sanders, 2003b, 2004). And, in both countries, the forecasting field expanded in its theoretical and modeling approaches as a result of conferences and journal symposia. In the UK, conferences were held at the University of Manchester in 2010 and at the London School of Economics in 2015, where scholars presented their ex ante general election forecasts. Further, the field was substantially expanded due to the efforts of Harold Clark, namely his editorship of *Electoral Studies* and his scholarly work with Sanders, Stewart, and

Whiteley. In particular, forecasts were published in special issues of *Electoral Studies* on the 2005, 2010 and 2015 elections. Moreover, the *British Journal of Politics and International Relations* ran a special issue on the 2005 parliamentary election. However, the snap elections in 2017 and 2019 hindered organizing conferences and journal symposia. In these elections, scholars turned to blogs and social media to issue their forecasts, which unfortunately means that today our record of these forecasts and the details of the modeling approaches used has become much more limited due to dead website links and the abbreviated nature of blog posts. In earlier years, when these forecasts were published in peer-reviewed journals, the articles contained rich explanations of modeling decisions, and they can be located in the journal archives.

In this article, we focus on the evolution of published model-based forecasts of UK parliamentary elections. As with most other democracies, the initial work examined the dependent variable of popularity, and eventually the vote itself. Often, these models are summarized as vote-functions (for a general discussion, see Stegmaier et al., 2017). We approach this chronologically, focusing first on the path-breaking early publications, and then by highlighting innovations in forecasting models. Throughout this tour of UK election prediction, we will see that scholars have tested different approaches to address challenges presented by the UK political context. These challenges include the fact that 1) it is the seat share, rather than the vote share, that determines which party "wins" the election; 2) there could be a

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“hung parliament” where no party wins a clear majority as has happened in 1974, 2010, 2017; and 3) the mostly two-party system has fragmented with the increased strength of the Liberal Democrats and regional parties. After reviewing the evolution of forecasting models, we then move on to assess their performance. And finally, we reflect on what the field can do to continue to move forward, while documenting our approaches and predictions.

## 2. Theoretical foundations and origins of UK election forecasting models

The earliest model-based election forecasts in both the US and the UK made use of popularity data (Sigelman 1979; Whiteley 1979) as the driving factor in their predictions. This approach stemmed from Mueller's (1970, 1973) research on US presidential approval and the Goodhart and Bhansali (1970) work on government support in the UK. Sigelman (1979) demonstrated how regression models based on a popularity time series could be used to predict how well an incumbent president running for re-election would do. In looking forward to the 1980 presidential context, he presents a range of scenarios to show what the vote share forecast would be using different popularity levels for President Jimmy Carter. In the same year, Whiteley (1979) published his model based on Labour and Conservative party popularity. His model applied a Box-Jenkins methodological strategy to generate vote and seat share forecasts for the two main parties for a hypothetical election in June/July 1978. In these two trail-blazing works, the authors focused more on demonstrating how the data and methodology could be used for prediction, and less on generating a final vote or seat share forecast for a real election. The timing of the submission, peer review, and publication process likely impacted Sigelman's decision to offer vote share forecasts based on different scenarios and Whiteley's choice to issue the prediction for a hypothetical election date when the actual election was held in May 1979. The length of time that the peer-review and publishing process takes at academic journals led some UK forecasters to adopt Sigelman's scenario strategy in order to get the forecast on record prior to the election taking place. Today, organized conferences, blogs, and well-timed journal symposia can better serve the time-sensitive nature of election forecasting.

The initial foray into structural-based modeling in the UK appears in Mughan's (1987) article that compared three different, yet basic, forecasting models. The first model, the “incremental model,” uses vote share in the previous election as the predictor for the next election. The second model, “the polling model,” uses vote intention for the focal party in the final Gallup poll released before the election, in order to forecast vote share. And the third model, the “economic model,” uses economic measures to predict election performance. The use of economic measures in a forecasting model – specifically the annual unemployment rate in the year before the election and Britain's GDP the year before the election – represents the first structural forecasting model in the UK, in that it introduces key factors that had been theoretically and empirically demonstrated to impact voting and election results through the economic voting and referendum theory literature.

Mughan (1987) estimates the three forecasting models on data covering 10 parliamentary elections, 1951–1983, to predict vote shares for the governing, opposition, and Alliance parties, and conducts a post-election assessment of the models' performance for the 1987 election. While all three models predict the Conservative party's electoral dominance in 1987, Mughan favors the economic model because it most accurately reflected the gap in support between the government and opposition at 11.5 percentage points which is identical to the actual results. To be more precise, the economic model produced point estimates of 39.0%, 27.5%, and 27.7% of the vote for the Conservative, Labour, and Alliance parties, respectively, in comparison to the actual results of 40.3%, 30.8%, and 22.6% of the vote. Additionally, for the time series, it has the best goodness of fit measures. Mughan's early work shows the utility of simple economic measures, such as

unemployment rate and GDP, for forecasting elections (though it did overestimate the third-party vote share).

## 3. 1992 Election

The 1992 UK parliamentary election marks the first in which an ex-ante model-based forecast was published in advance of the election. Sanders (1991, 248–249) analyzed how monthly party support moved with economic conditions and perceptions over the post-war period. Starting in 1987, he observed that inflation and interest rates were linked to personal economic expectations and that these measures could be used to explain party popularity. He argues that this relatively short time span – covering just one parliamentary term – would capture the dynamics of support for the incumbent government and the surrounding political environment. Sanders' approach is based on two equations. First, he forecasts personal economic expectations from interest rates, inflation, and a poll tax dummy. Then he moves on to predict Conservative party popularity from popularity (t-1), interest rates (t-1), personal economic expectations (t-3) and poll tax dummy over the period of February 1987 to October 1990. For the expected 1992 elections, he next provides popularity prediction scenarios based on different levels of inflation and interest rates. Then, to issue the forecast for the expected 1992 election, he uses a similar approach to Sigelman (1979) in that his predictions are based on different levels of inflation and interest rates. An important difference between the Sigelman (1979) and Sanders (1991) scenarios is that in the US, presidential election timing is fixed, so Sigelman knew exactly when the next election would take place, while in the UK, elections can be called earlier than the end of the 5-year parliament term.

One of Sanders's (1991) scenarios, with 10% interest rates and 5% inflation, predicted the Tories would have 42.5% popularity/vote share in 1992. His forecast was nearly spot on and “far more accurate than forecasts, based on exit polls, that were made on election night itself” (Sanders 1995, 255).

Another publication, this one released well after the 1992 election, by Clarke and Stewart (1995), further developed the structural-model approach. They generate a vector-autoregression (VAR) model of Conservative party support that is based on previous party support, national economic expectations, and a “misery index” (combination of unemployment and inflation) using data from 1979 to 1991. They determine that this model, which combines forward-looking economic expectations and current conditions, performed better in predicting the April 1992 election than other VAR models they estimated.

## 4. 1997 Election

Few forecasters attempted to predict the outcome of the 1997 general election. We speculate that the lack of interest could be for two reasons: 1) low level of competition; and 2) UK forecasting had not yet become a systematic enterprise. In the 1997 election, the Labour Party was the clear favorite to win by a large margin, so interest in forecasting may have been low due to the lack of competition. Alternatively, or potentially compounding the lack of competitiveness, is the relative youth of the forecasting field in the UK at that time. The development of forecasting teams and consistent access to peer-reviewed journal symposia did not come about until the 2005 election. For one, or both, of these reasons, only Sanders (1995) developed a forecast for this election.

Sanders' (1995) forecasting model for the 1997 election builds upon his earlier model (Sanders 1991) predicting government popularity, but he modifies the specification to include taxation and unemployment. The monthly time series for estimating the model spans from July 1987 to December 1993. Again, Sanders estimates predictions for the next election based on different scenarios. Ultimately, he concludes that the Conservative government would be unlikely to win, based on the previous strategy of reducing interest rates, largely because the interest rates were already quite low; however, lowering the tax rate would lead

to a better chance of Tory victory. While no other models were introduced to forecast this contest, it is worth noting that innovation was still occurring, in that Sanders' effort involved changing the specification of an established model, in order to improve accuracy.

## 5. 2001 Election

Lewis-Beck et al. (2004) develop a structural forecasting model partially derived from Lewis-Beck's earlier work on US elections (Lewis-Beck and Rice, 1984a; and b, 1992; Lewis-Beck and Mark Wrighton, 1994; Lewis-Beck and Tien, 1996). The forecast was issued in advance of the 2001 election at the Public Choice Society meetings in San Antonio in March 2001 and later published in *Electoral Studies*. The model predicts incumbent party vote share and is based on elections covering 1955 to 1997 ( $N = 12$ ) with a lead time of 6 months. The model is based on political and economic measures including inflation, public approval of the government's record, and the number of terms the governing party has been in office. This "terms in office" measure captures the tendency for democratic publics to want leadership change after some time has passed.

The Lewis-Beck et al. (2004) forecast for an anticipated April 2001 election was that Labour's vote share would be 46.5%, which they translate to a strong parliamentary majority of 69% of the parliamentary seats. The 2001 general election was actually held two months later, in June 2001. Labour won the election with 40.7% vote share and a comfortable majority of 62.5% of the seats. Thus, even though the election took place slightly later than Lewis-Beck et al. expected, their model correctly predicted a Labour victory.

## 6. 2005 Election

In modern democracies, we can observe cycles alternating between parties on the Left and the Right winning elections. This idea underpins the "cost of ruling" or "terms in office" measures sometimes included in forecasting models (Lewis-Beck et al., 2004) and forms the basis of Norpoth's (2004) forecasting approach for the 2005 election. He develops an auto-regressive moving average (ARMA) model, covering elections 1831–2001 ( $N = 35$ ) based on previous Conservative Party share of the major (2 party) vote. Since the model requires just the previous vote share, it has a very long lead time. Essentially, once the election results are in, it can offer a prediction for the next election. For the 2005 election, the model's two-party vote share forecast was Conservatives 45.9%; and Labour 54.1%. In the election, the Conservatives performed somewhat better than the prediction with 47.9% of the two-party vote share; and Labour received 52.1%.

Whiteley (2005) takes on the challenge of directly predicting party seat shares, since it is seat share, rather than vote share, that ultimately determines the winner. Whiteley applies the "Cube Rule" in his "votes to seats" model. The data used in the 2005 prediction span elections from 1945 to 2001 ( $N = 16$ ), and the model is estimated with a 6-month lead time, in anticipation of an expected May 2005 election (which is in fact when the election occurred). The model predicts party seat shares based on the party's seat share in the previous election and the vote share of the Conservative, Labour, and Liberal Democratic parties in the polls 6 months in advance of Election Day. In order to issue a forecast based on these models, assumptions had to be made about the number of seats that minor parties would win, since minor parties are not considered in the prediction model. Whiteley assumes minor parties will win 28 seats, which is based on the 2001 minor party seat share in addition to a reduction for Scotland. Ultimately, Whiteley's model predicted that Labour would win 358 seats, Conservatives 213, and Lib Dems 47. This forecast rather accurately predicted the actual result, especially for Labour. Labour won 355 seats, the Conservatives 198 seats, and Lib Dems 62 seats.

## 7. 2010 Election

In 2010, many of the forecasts cited were included in a special issue of *Electoral Studies*. This election, with the prospect of a hung parliament, a rare event that has only occurred three times in the past century, posed a special challenge for forecasters. To their good fortune, almost all correctly predict this outcome. Whiteley (2008) generates and tests rival models, including a Seats-Votes model and an Economic model (with predictors of inflation, government approval, and terms in office), both estimated with a 6-month lead. The Seats-Votes model had a better record for the Tories and Liberal Democrats, but the combined model did best for Labour. Regardless, they predicted a hung parliament. A related effort, by Whiteley et al. (2011), further develops the Seats-Votes model, drawing on Whiteley's (2005) measures. Again, the prediction is for a hung parliament. Lebo and Norpoth (2011), in a contemporary but somewhat different approach, formulate an autoregressive vote-seat translation, which also foretells a hung parliament.

Changing the focus from the votes-seats swing, Lewis-Beck et al. (2011) introduce the innovation of "nowcasting," which generates forecasts over the election cycle, making new predictions every three months. Their "two-step model" first estimates incumbent approval as a function of PM and Government support, inflation, interest rates, and terms in office (Nadeau et al., 2009). Then, second, it estimates vote share as a function of this incumbent approval variable. Once more, the forecast was for a hung parliament. Fisher et al. (2011), in another two-step process, first estimate vote share from aggregated vote intention polls, then with that result estimate seat share, also predicting a hung parliament.

Still other innovations are introduced, in this fruitful competition. Rallings et al. (2011) base their model, exceptionally, on local by-election results. This approach yields yet another forecast of a hung parliament. Not merely an echo, Lewis-Beck and Stegmaier (2011), take their first foray using citizen forecasting (CF) to predict this UK election. To clarify, CF relies on voter expectations, rather than vote intentions (as traditional polling employs). This effort stands out because of its methodological uniqueness, and because it forecasts a Conservative majority, unlike the rest of the pack.

## 8. 2015 Election

The 2015 election shows a strong, continuing interest in election forecasting in the UK, culminating in a large special issue in *Electoral Studies* (2016). Familiar names from the 2010 contest roll out versions of their approach, namely Whiteley et al. (2016) with a Seats-Votes model, Lebo and Norpoth (2016) with a PM and Pendulum Model and Fisher and Lewis-Beck (2016) with a probabilistic model relying on aggregated polling data. Ford et al. (2016) offer an aggregated polling data-based model (following the same approach as Fisher et al., 2011).

There are also new names associated with new perspectives. Burnap et al. (2016) present a twitter-based model. Rallings et al. (2016) forecast using council election results, as well as local by-elections. Prosser (2016) also deploys local elections results, but in a different way, i.e., the party's share of the national vote in local elections. Mellon and Fieldhouse (2016) depart from actual election results, in a turn to forecasts based on British Election Study internet panel data. Hanretty et al. (2016) also depend on polling data, but at the constituency level as well as the national level. Murr (2016) brings back citizen forecasting by applying his constituency-level model that he tested *ex post* on the 2010 election (Murr 2011).

In addition to innovations in data, 2015 heralds innovation in the specification of more traditional political economy-type models. Lewis-Beck et al. (2016) unveil the synthetic model, which blends the classic structural equation (e.g., government popularity and economic growth as predictors), with the polling predictor variable of vote intention. Stegmaier and Williams (2016) return to the original dependent variable in the UK forecasting run, that of popularity. They specify this current

popularity model with a new mix of independent variables (i.e., partisanship, retrospective economic and PM evaluations, political conditions and events) and estimate it with a hitherto untried estimation technique, seemingly unrelated regression, SUR. (This technique was later applied again to the UK case by Mongrain, 2021). In many ways, 2015 serves as the banner year for growth and experimentation in UK election forecasting, including a turn to blog posts (e.g., Whiteley and Clarke, 2015).

Although it was a banner year, it was also riddled with disappointment for forecasters, pollsters, and the media because of their collective failure to predict a Conservative majority. None of the forecasting teams correctly predicted the Conservative majority, though many predicted a Conservative plurality, with Murr's (2016) citizen forecast performing best, predicting a 30-seat spread in favor of the Conservative Party. However, the election resulted in a 98-seat spread in favor of the Conservative Party.

The disappointing predictions from forecasters, who were so triumphant only five years prior, had several implications for the field. First, many models used vote intention polls that were flawed, leading to poor forecasts and a reevaluation of opinion polling in the UK. The UK developed a commission to investigate the issues related to opinion polling in the 2015 election, and other polling companies, such as YouGov, also reevaluated their polling techniques. Second, the 2015 election was the beginning of a trend toward a multiparty system in the UK, which has made it difficult to get a majority in the House of Commons because of the rise of regional parties such as the SNP.

## 9. 2017 and 2019 Elections

Both these elections were 'snap,' called rather suddenly and relatively close to other elections on the calendar. About two years out from the 2015 contest, the June 2017 election was announced, with Prime Minister Theresa May hoping to secure a larger majority, in order to facilitate Brexit negotiations. Just two and one-half years later, Prime Minister Boris Johnson called the December 2019 snap election, again hoping for greater parliamentary support behind his Conservative plan to withdraw the UK from the EU. Unlike other elections, these contests offered little lead in the run up to the ballot, making it difficult for forecasters, in timely fashion, to formulate their models, gather necessary data, write up the analysis, and submit the work to a journal for peer-review. Instead, they took the more efficient, if less curated, route of online appearances.

For 2017, Stephen Fisher, John Kenny and Rosalind Shorrocks (June 8, 2017), online, stated "Our final combined forecast is for a Conservative majority (sic) 66 ... well down on the 132 majority we first forecast on 12th May." Tweeting on April 18, 2017, Matthew Lebo reported a Conservative lead over Labour of 169.5 seats, based on 100,000 simulations. Coates (2017), in a *Times* piece, predicted "shock losses" for the Conservatives. In *Medium*, Janta-Lipinski (2017), discussed "yet another election prediction." In the *Monkey Cage* (June 6, 2017), Murr, Stegmaier, and Lewis-Beck, again applying the citizen forecasting method, foresaw that the Conservatives were likely to gain a controlling majority, with a net seat gain of 31 seats. With respect to the 2019 snap election, Murr, Stegmaier, and Lewis-Beck (LSE Blog, December 9, 2019) once more used citizen forecasting to predict a big Conservative win. They called the Conservatives with 360 seats, compared to only 190 for Labour. Based on their assessment of voter expectations (rather than voter intentions), these forecasts turned out to be highly accurate, as the actual seat shares were Conservatives: 365, Labour 202 (Lib Dems 11, SNP: 48.) Citizen forecasting, as a relatively new forecasting strategy, continues to hold promise. In addition to its claims of accuracy, it is low cost, simple, and quick, compared to other methods. It is important to note, though, that while we emphasize citizen forecasting in this section, other models were used to predict the outcome of the 2017 and 2019 snap elections. However, because of the short notice of these elections, many forecasters published their forecasts through online forums (e.g.,

social media, blog posts, etc.) that did not provide the requisite details to assess these models. The question of accuracy, as well as other desirable forecasting traits, are evaluated more fully below.

## 10. Evaluating the forecasts: accuracy, lead, and other desirable properties

With the possible exception of the US, the UK represents the most forecast electorate in the democratic world. It behooves us, therefore, to consider the worth of these forecasts. Accuracy appears to be a paramount criterion. This is evident from a leading quality index for forecasting instruments, which weights accuracy by a factor of three, when compared to other factors (Lewis-Beck 1985, 60; see also Lewis-Beck and Rice 1992; Lewis-Beck 2005). As a leading economist, Arnold Zellner, noted, "Prediction is central in science and in evaluating alternative generalizations or models" (García-Ferrer, 1998, pg 309). In other words, election forecasts can indicate how well our election theories are working. Also, they may serve a practical political purpose, suggesting how our favorite (or least favorite) candidates or parties are likely to do at the ballot box. Strictly speaking, forecasting is not always easy, because it involves making predictions *ex ante*, i.e., about events that have yet to occur. Since such forecasts are necessarily uncertain, the question of their performance, beyond chance guessing, merits pursuit. Hence, for this UK case, we examine scientific estimates based on independent variables measured and available before the election. In that pursuit, we offer an "accuracy grid" in Table 1 for *ex ante* UK election forecasts reviewed here, focusing on point estimates (how close is the predicted vote share to the actual vote share?) and on general outcome, (is the winner correctly predicted?).

As important as accuracy is, there are other criteria, namely lead, parsimony and reproducibility (Lewis-Beck, 2005). Lead refers to the distance in time from the forecast and to the election itself. The greater the distance from the contest, the more challenging the accuracy standard. In general, the closer to the election the better; however, when the forecast takes place on the heels of the election, it holds much less interest, for the result will be known in a few days. Recent work has shown that a lead time of around three months tends to optimize accuracy (Jennings et al., 2020).

Following the principle of Ockham's razor, a forecast should be parsimonious, in the sense of limiting the number of predictor variables and the number of predictions made. Having a large number of independent variables can not only be confusing; it can also capitalize on chance fitting that exaggerates the power of a model. Relatedly, forecasters who release forecasts repeatedly risk undercutting their credibility. Consider the analogy from a dart game. We value more the player who uses fewer darts and does not stand too close to the dart board. In fact, as we know, standing face-to-face with the dartboard can yield more error, as illustrated in the October polling of the Clinton 2016 race in the US (Jennings et al., 2020). The standard of parsimony encourages achievement of the fourth criterion, reproducibility. The more complicated the model design, specification, and measurement, the harder it becomes for other forecasters to reproduce, and thus validate (or not), the researchers work. If measures are costly—in terms of time, money, access, or understanding—then its scientific credibility diminishes. Happily, most serious scientific forecasters follow high standards of transparency in reporting, so enabling others to assess their work.

Table 1 reports the main predictors and accuracy of UK election forecasts from their inception in 1979 to, most recently, 2019. We evaluate each election forecast from 1979 to 2019 in terms of accuracy. Each forecast is assessed in relative and absolute terms. In other words, we evaluate forecasts based on whether they predicted the outcome correctly (e.g., Labour majority, Conservative majority, hung parliament, etc.) and if the forecast is close in its point estimate. There is some debate about which measure of accuracy we should use, but our view is that it is important to predict elections correctly in the relative and absolute sense (Mughan 2004). In Table 1, we provide the relative

**Table 1**  
UK election forecasting models: Predictors and accuracy.

| Authors                                                         | Main Predictors                                                                                                  | Forecast Prediction (votes)             | Forecast Prediction (seats)       | Election Results <sup>a</sup>         | Correctly Predicted <sup>b</sup> |
|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-----------------------------------|---------------------------------------|----------------------------------|
| <b>1979 General Election: Conservative Party won a majority</b> |                                                                                                                  |                                         |                                   |                                       |                                  |
| Whiteley (1979)                                                 | Popularity                                                                                                       | Labour: 43%<br>Conservatives: 45.6%     | Labour: 275<br>Conservatives: 319 | Labour: 269<br>Conservatives: 339     | Yes                              |
| <b>1987 General Election: Conservative Party won a majority</b> |                                                                                                                  |                                         |                                   |                                       |                                  |
| Mughan (1987)                                                   | Unemployment; GDP                                                                                                | Labour: 27.5%<br>Conservatives: 39%     | N/A                               | Labour: 30.8%<br>Conservatives: 42.3% | Yes                              |
| <b>1992 General Election: Conservative Party won a majority</b> |                                                                                                                  |                                         |                                   |                                       |                                  |
| Sanders (1991)                                                  | Popularity; Economic expectations (individual); Inflation; Unemployment; Interest rate                           | Conservatives: 42.5%                    | N/A                               | Conservatives: 41.9%                  | Yes                              |
| <b>1997 General Election: Labour Party won a majority</b>       |                                                                                                                  |                                         |                                   |                                       |                                  |
| Sanders (1995)                                                  | Popularity; Economic expectations (individual); Taxation; Unemployment                                           | Conservatives: 37.5%                    | N/A                               | Conservatives: 30.7%                  | Yes                              |
| <b>2001 General Election: Labour Party won a majority</b>       |                                                                                                                  |                                         |                                   |                                       |                                  |
| Lewis-Beck et al. (2004)                                        | Inflation; Government approval; Cost of ruling                                                                   | Labour: 46.5%                           | Labour: 69%                       | Labour: 402 (62.2% of the seats)      | Yes                              |
| <b>2005 General Election: Labour Party won a majority</b>       |                                                                                                                  |                                         |                                   |                                       |                                  |
| Bélanger et al. (2005)                                          | Inflation; Government approval; Cost of ruling; Difference in vote intention for Labour and Conservative parties | Labour: 41.8%                           | Labour: 57.7%                     | Labour: 355 (55% of the seats)        | Yes                              |
| Norpoth (2004)                                                  | Conservative vote share                                                                                          | Labour: 54.1%<br>Conservatives: 45.9%   | N/A                               | Labour: 35.2%<br>Conservatives: 32.4% | Yes                              |
| Sanders (2005b)                                                 | Government popularity (party and PM); Economic expectations (individual); Cost of ruling                         | Labour: 39%<br>Conservatives: 31%       | N/A                               | Labour: 35.2%<br>Conservatives: 32.4% | Yes                              |
| Sanders (2005a)                                                 | Government popularity (party and PM); Economic expectations (individual); Cost of ruling                         | Labour: 38%<br>Conservatives: 31%       | N/A                               | Labour: 35.2%<br>Conservatives: 32.4% | Yes                              |
| Whiteley (2005)                                                 | Seat share; Vote share                                                                                           | N/A                                     | Labour: 358<br>Conservatives: 213 | Labour: 355<br>Conservatives: 198     | Yes                              |
| <b>2010 General Election: Hung Parliament</b>                   |                                                                                                                  |                                         |                                   |                                       |                                  |
| Fisher et al. (2011)                                            | Estimated vote intention from opinion polls                                                                      | Labour: 32.6%<br>Conservatives: 39.3%   | Labour: 275<br>Conservatives: 299 | Labour: 258<br>Conservatives: 306     | Yes                              |
| Lebo and Norpoth (2011)                                         | PM approval                                                                                                      | N/A                                     | Labour: 287<br>Conservatives: 285 | Labour: 258<br>Conservatives: 306     | Yes                              |
| Lewis-Beck and Stegmaier (2011)                                 | Vote expectations                                                                                                | N/A                                     | Conservatives: 358                | Conservatives: 306                    | No                               |
| Lewis-Beck et al. (2011)                                        | Proxy for incumbent government approval                                                                          | Labour: 36%<br>Conservatives: 42%       | Labour: 303<br>Conservatives: 293 | Labour: 258<br>Conservatives: 306     | Yes                              |
| Rallings et al. (2011)                                          | Local by-election results                                                                                        | Labour: 27.2%<br>Conservatives: 35.7%   | Labour: 237<br>Conservatives: 299 | Labour: 258<br>Conservatives: 306     | Yes                              |
| Whiteley (2008)                                                 | Seat share; Vote intention; Government approval; Inflation                                                       | N/A                                     | Hung parliament                   | Labour: 258<br>Conservatives: 306     | Yes                              |
| Whiteley et al. (2011)                                          | Seat share; Vote intention                                                                                       | N/A                                     | Labour: 273<br>Conservatives: 282 | Labour: 258<br>Conservatives: 306     | Yes                              |
| <b>2015 General Election: Conservative Party won a majority</b> |                                                                                                                  |                                         |                                   |                                       |                                  |
| Fisher (2016)                                                   | Aggregated vote intention polls                                                                                  | Labour: 32%<br>Conservatives: 35%       | Labour: 262<br>Conservatives: 285 | Labour: 232<br>Conservatives: 330     | No                               |
| Ford et al. (2016)                                              | Vote intentions                                                                                                  | Labour: 32.6%<br>Conservatives: 35%     | Labour: 272<br>Conservatives: 274 | Labour: 232<br>Conservatives: 330     | No                               |
| Hanretty et al. (2016)                                          | Swing model                                                                                                      | Labour: 32.8%<br>Conservatives: 34.4%   | Labour: 267<br>Conservatives: 278 | Labour: 232<br>Conservatives: 330     | No                               |
| Lebo and Norpoth (2016)                                         | Coalition adjusted PM approval; number of terms in office, adjusted for SNP power                                | N/A                                     | Labour: 263<br>Conservatives: 287 | Labour: 232<br>Conservatives: 330     | No                               |
| Lewis-Beck et al. (2016)                                        | Government popularity; GDP growth; median vote intention                                                         | Labour: 31.8%<br>Conservatives: 34.4    | Labour: 280<br>Conservatives: 290 | Labour: 232<br>Conservatives: 330     | No                               |
| Mellon and Fieldhouse (2016)                                    | Destination vote share 2010                                                                                      | N/A                                     | Labour: 278<br>Conservatives: 274 | Labour: 232<br>Conservatives: 330     | No                               |
| Murr (2016)                                                     | Citizen forecast                                                                                                 | N/A                                     | Labour: 262<br>Conservatives: 292 | Labour: 232<br>Conservatives: 330     | No                               |
| Prosser (2016)                                                  | Party's share of the national vote at the local election; dummies for Labour, Lib Dem, other, and incumbent      | Labour: 31.05%<br>Conservatives: 35.25% | Labour: 287<br>Conservatives: 296 | Labour: 232<br>Conservatives: 330     | No                               |
| Rallings et al. (2016)                                          | Local by-election results                                                                                        | Labour: 33%<br>Conservatives: 33%       | Labour: 280<br>Conservatives: 275 | Labour: 232<br>Conservatives: 330     | No                               |
| Stegmaier and Williams (2016)                                   | Party's vote share <sub>t-1</sub> ; PID; retrospective economic evaluations; party's position in government      | Labour: 34.75%<br>Conservatives: 31.13% | N/A                               | Labour: 30.4%<br>Conservatives: 36.8% | No                               |
| Whiteley, Clarke, Sanders, and Stewart (2016)                   | Party's seat share <sub>t-1</sub> ; Labour and Conservative predicted vote share 6 months prior to election      | N/A                                     | Labour: 281<br>Conservatives: 271 | Labour: 232<br>Conservatives: 330     | No                               |
| <b>2017 General Election: Hung Parliament</b>                   |                                                                                                                  |                                         |                                   |                                       |                                  |

(continued on next page)

Table 1 (continued)

| Authors                                                         | Main Predictors   | Forecast Prediction (votes) | Forecast Prediction (seats)       | Election Results <sup>a</sup>     | Correctly Predicted <sup>b</sup> |
|-----------------------------------------------------------------|-------------------|-----------------------------|-----------------------------------|-----------------------------------|----------------------------------|
| Murr et al. (2017)                                              | Vote expectations | N/A                         | Conservatives: 361                | Conservatives: 317                | No                               |
| <b>2019 General Election: Conservative Party won a majority</b> |                   |                             |                                   |                                   |                                  |
| Murr et al. (2019)                                              | Vote expectations | N/A                         | Labour: 190<br>Conservatives: 360 | Labour: 202<br>Conservatives: 365 | Yes                              |

<sup>a</sup> Election results are reported as the number of seats the parties received in the election unless the forecast predicted vote share only. Then we report the vote share received by the parties.

<sup>b</sup> Correctly Predicted represents whether the model correctly predicted the majority party or hung parliament.

accuracy with a simple right/wrong measure, but we also report the point estimate of each forecast and compare these to the actual outcome. Generally speaking, forecasters have been fairly accurate in relative terms, excepting the 2015 forecasts, but less accurate in their point predictions.

Early forecasts were quite accurate in relative and absolute terms. Whiteley's (1979) predictive model for a hypothetical 1978 UK election, reported in the table as a 1979 forecast, is quite accurate given that its prediction was given for a potential earlier election. He predicted the Conservative majority within 20 seats and the opposition within 6 seats. The predictive accuracy of this forecast is quite commendable given the likely constraints in the peer-review process described above. Shifting to the first explicitly *ex ante* forecasts in the UK, Sanders (1991) provides an amazingly accurate forecast of popularity/vote share for the Conservative Party. The Conservative Party was predicted to win 42.5% of the vote given the conditions of an interest rate of 10% and an inflation rate of 5%. The Conservative Party won 41.9% of the vote in the 1992 election. The prediction underestimates vote share by less than 1 percentage point. This is an incredible prediction and provides further merit to simple economic forecasting models. Building on this work, Sanders forecasts a hypothetical 1996 election using a similar economic model. This forecast correctly predicts the Conservative Party's 1997 loss to the Labour Party. In terms of relative accuracy, the model is not as accurate, with a 6.8 percentage point prediction error. However, this is due to Sanders (1995) setting up unrealistic scenarios to show that the chances of a Conservative victory in the election were extremely unlikely.

The accuracy of these forecasts in absolute terms declined in 2001 with Lewis-Beck, Nadeau, and Belanger's (2004) predictive economic models. The forecast target is the governing Labour Party. They correctly predict a Labour majority in 2001. Though, in absolute terms, Lewis-Beck, Nadeau, and Belanger predicted the Labour Party would receive 46.5% of the vote, which is a 5.8 percentage point overestimation of the actual vote. Furthermore, they predicted the Labour Party would receive 62% of the seats, but the actual percentage proved to be 7 percentage points higher than their economic model predicted six months earlier. As Mughan (2004) clearly points out, the inaccuracy of this forecast may partially be caused by the "foot-and-mouth endemic" in 2001, but more concerningly, by model misspecification. He makes three recommendations for future economic models: using change in inflation instead of inflation rate, changing the dependent variable to share of the two-party vote, and exploring expected turnout measures. These modifications to the economic model may improve the absolute accuracy of the economic model in future elections.

*Electoral Studies* and the *British Journal of Politics and International Relations* each published special issues with forecasts of the 2005 election where many of the forecasters predicted the election outcome well in both absolute and relative terms. In this election, the actual outcome was a Labour Party majority, where they received 35.2% of the vote. All the forecasts correctly predict a Labour majority or the vote share for the Labour, Conservatives, and Liberal Democrats within their confidence intervals. Of particular note, Whiteley correctly predicts the Labour majority within 3 seats; Bélanger et al. (2005) predict the percentage of Labour seats within 2.2 percentage points; and Sanders (2005) predicts the percentage of the three major party vote shares within his confidence

intervals. Overall, economic models and the vote-seat model did quite well to predict the 2005 general election in an absolute and relative sense.

In the 2010 election, we saw much innovation and success from a diverse set of forecasting teams. The success of forecasters is especially outstanding considering that the outcome was a hung parliament, a rare event that had only occurred three times in the past century (Gibson and Lewis-Beck 2011). On the whole, all but one forecast, Lewis-Beck and Stegmaier (2011), predicted a hung parliament, where no party received the majority of seats in the lower chamber. Thus, there was a high degree of relative accuracy across these forecasts. Furthermore, the forecasts were quite accurate in absolute terms as well. The closest prediction for the governing Labour Party's seat total is provided by Whiteley et al.'s (2011) impressive "Seats-Votes" model, where they predicted the outcome within 15 seats. The opposition Conservative Party received 306 seats. The closest predictions come from Fisher et al. (2011) and Rallings et al. (2011), who each predicted the Conservative Party to receive 299 seats, which is an underestimation of only 7 seats. The only failed forecast predicted a Conservative majority. This forecast from Lewis-Beck and Stegmaier (2011) uses a citizen forecasting model, which uses vote expectations as the primary predictor of vote share. Though unsuccessful, the citizen forecasting model holds great utility as shown by Murr's (2011) *ex post* forecast. Using citizen forecasting to predict constituency winners, he predicted the Labour Party's actual seat total within 2 seats. Though predicting elections is, of course, easier *ex post*, this shows the potential of the parsimonious, cost-, and time-efficient citizen forecasting model. Overall, these forecasts were quite accurate, especially after accounting for the considerable lag time of many of them.

After much success from the UK forecasting community over the previous three elections, the 2015 election provided an unanticipated outcome. In 2015, the Conservative Party won a majority of the seats in parliament. This was much to the surprise of forecasters. None of the forecasters predicted the outcome correctly. The most predicted outcome was a repeat of the 2010 election: a hung parliament. All the forecasts underestimated the seats received by the Conservative Party. The closest forecast of Conservative Party seats is provided by Prosser (2016), who predicted that they would receive 296 seats. This is an underestimation of 34 seats. Although these forecasts failed to predict the two major party seat shares, those that predicted seats for the Scottish Nationalist Party (SNP) did well. Fisher (2016), Ford et al. (2016), and Hanretty et al. (2016) correctly predicted SNP seats within 3 seats. Other forecasters also did well to predict the unprecedented surge in UKIP vote share. However, despite the prediction of smaller party vote- and seat-shares, forecasters did quite poorly in this election. It was not just the forecasters, however, but also journalists and pollsters who incorrectly predicted the election (Hanretty and Jennings 2015).

The 2017 and 2019 UK general elections were, as described above, snap elections, and therefore, we have few peer-reviewed forecasts due to the lack of lead time. Murr et al. (2017) provide a "citizen forecast" that predicts the governing Conservative Party to receive 361 seats. This forecast overestimated the Conservative Party's success by 44 seats and failed to predict a hung parliament. Other forecasting teams, such as Fisher et al. (2017) and Lebo and Norpoth. (2016), similarly predicted a

Conservative majority that did not come to fruition. In 2019, the forecasting team of Murr et al. (2019) again produced a citizen forecast. The forecast predicts Labour and Conservative Party seats. The Labour Party received 202 seats, while the Conservative Party received 365 seats in this election. Murr et al. (2019) predicted that the Labour and Conservative Parties would receive 190 and 360 seats, respectively. The small absolute error is quite the feat considering the context of the snap election. This continues to show the utility of citizen forecasting in predicting election outcomes (see also Murr et al., 2021).

## 11. Lessons and concluding thoughts

The current enterprise of election forecasting in the UK grew from the seeds planted in the 1970s and 1980s, as manifest particularly in the signpost research of Whiteley (1979) and Mughan (1987). The former designated the relevant dependent variable to be prediction of the popularity of the two leading parties, the Conservatives and Labour, as determined by macroeconomic conditions and the electoral cycle. This choice led many subsequent researchers down the path of discovering the optimal popularity function, as we have seen. This approach can be said to have crested in 2015, as represented in the paper of Whiteley et al. (2016). The vote function approach also appears to have reached its high-water mark in 2015, with the political economy model of Lewis-Beck et al. (2016), in their ‘synthetic’ equation, which maintained Mughan’s format for forecasting incumbent vote share as a function of the economy and popularity (but measured as vote intention).

After 2015 the diaspora began, with newer forecasting approaches taking a firmer hold, e.g. aggregated vote intention models, citizen forecasting, and combined models perhaps using local data. This more innovate work faced more challenges because it had to deal with two snap elections in a row, the vagaries induced by Brexit, and the corollary blossoming of a more multi-party system. As well, the possible venues for pre-election publication changed, relying much less on peer-reviewed scholarly journals and much more on social media outlets. This change created opportunities for more rapid and widespread dissemination of forecasts. Online blogs, tweets, and websites allow for immediate public release of forecasts. However, they often lack theoretical and methodological details of the models, which are necessary to validate and build on the work. The lack of detail in online forums inhibits our ability to replicate forecasts; thus, these necessarily rate low on reproducibility (Lewis-Beck 2005). Moreover, some links to known online forecasts are now dead, and so rest in a ghost’s graveyard. This lack of transparency and the temporary nature of online forums could lead to increased skepticism, especially among seasoned forecasters, but some blog outlets do have better reputations than others for publishing forecasts because of their efforts to thoroughly report theoretical and methodological details. This raises the demand for placing early election forecasts in a journal, or other archivable outlets, for that purpose.

In the evolution of election forecasting in the UK, we see the stamp of Harold Clarke virtually everywhere, in the literature and in public advocacy, stretching back to his earliest collaborations—with Whiteley, Sanders, and Stewart—and reaching to later collaborations, including publication efforts, such as his extended editorship of *Electoral Studies*, where he was attentive to the problems and prospects facing the study of UK elections and their forecasting. His is a banner name flying high over these fertile fields.

## Declaration of competing interest

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## Data availability

No data was used for the research described in the article.

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